

WHAT REMAINS?

SoHo + surrounding streets | Street-tree survival, 2016–2036

01 What I am attempting

I want to ask how many of the street trees recorded in my SoHo study area could remain in 2036, and where losing a few trees could leave a grid cell with none of its original living trees. My last map counted what was there; this sketch asks how that pattern could change over twenty years. I would test a **no-replacement scenario**, not predict that the city will stop planting.

02 The dataset

NYC Parks' **2015 Street Tree Census** was collected by staff, volunteers and partners from **May 2015 to October 2016**. Field records of tree locations, species, diameter and condition supported inventory and asset management. [1] My starting subset is **1,343 trees marked Alive** in the SoHo-centered study window—the count shown on my Assignment 2 map.

SOURCE COLUMN(S)	WHAT I WOULD KEEP	UNITS
tree_id	Identify the same original tree through time	Identifier; no units
status	Select records marked Alive	Category
longitude, latitude	Keep each tree at its recorded location	Decimal degrees
spc_common; tree_dbh	Retain baseline species and trunk diameter	Name; inches [4]

Species and diameter remain descriptive fields in this first sketch; they do not change the loss probability.

03 Research

Roman, Battles and McBride (2016, p. 9, eq. 10) relate annual survival to the proportion of an original population remaining over time. [2] Hilbert et al. (2019) report median annual mortality of **2.3–2.6%** for repeated inventories of mixed-age urban trees. [3] I would adapt that approach to tree-by-tree yearly steps, as in Tutorial 3, but track survival rather than tree growth or ecological benefits. [4]

Population check: expected survivors = $1,343 \times (1 - m)^t$

Derived from [2]; m = annual loss probability and t = years after the 2016 baseline.

04 My record

One row = one original living tree. I would treat the 2015–2016 observations as a single **2016 baseline** (my simplifying assumption), add **present = 1**, and assign a cell ID using my existing **450 × 450-foot grid**. Coordinates stay fixed; mapping uses **NAD83 / New York Long Island (ftUS), EPSG:2263**, as in Assignment 2.

05 My rule

At each annual step, every still-present tree has probability **m** of dying or being removed; otherwise it remains. A lost tree stays at **present = 0** in all later years, and no replacement is added.

1.6% / YEAR

Lower-loss test

Published first quartile [3]

2.5% / YEAR

Central test

My choice within median range [3]

3.3% / YEAR

Higher-loss test

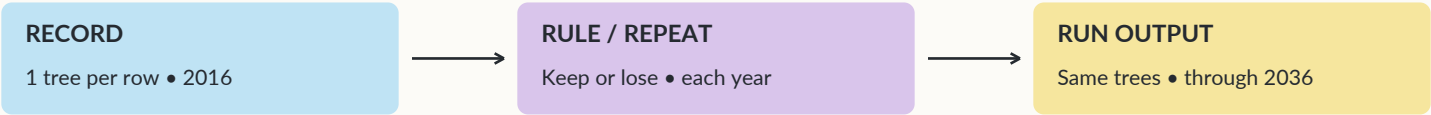
Upper end of third quartile [3]

Least certain: borrowing these rates for SoHo and holding one rate constant across all trees and years. Independent losses are also my assumption; none of these rates was measured in my study area.

THE RUN, BEFORE THE RUN

06 My run

2016 → 2036, one year at a time: 20 updates and 21 annual snapshots. I would repeat each rate scenario **100 times** (my initial choice), drawing each still-present tree’s loss independently with probability *m*. Only which trees are lost is random. All post-baseline years are modeled, including years that have already passed; this is not a verified present-day inventory.



07 The data it would generate

One row per tree, year, run and scenario, retaining its location and cell ID. New fields are marked * below. **The rows are invented to explain the structure—not observed data or simulated results.**

tree_id	cell_id	scenario*	run_id*	year*	present*	new_loss*
DEMO-A	A	2.5%	1	2016	1	0
DEMO-A	A	2.5%	1	2017	1	0
DEMO-A	A	2.5%	1	2018	0	1
DEMO-A	A	2.5%	1	2019	0	0

present: 1 = original tree alive and at its site; 0 = dead or removed. new_loss: 1 only in the loss year. Year is a calendar year; scenario is an annual probability; run ID is a label.

MAP SKETCH / TWO HYPOTHETICAL CELLS, NOT ACTUAL SOHO LOCATIONS



The actual map would use all 80 original grid cells, with per-cell survivor counts and shares. The 0% and 50% display breaks are my choices; cells empty at baseline stay unfilled. Across runs, I would also count how often each initially occupied cell loses all its original living trees.

08 What it cannot see

- New planting and replacement trees: this follows only the original recorded population.
- Species, size, care and site-specific risks, or shared shocks such as storms; extra runs do not fix a wrong loss rule.
- Canopy, shade, cooling or noise reduction: living-tree counts are not measurements of those effects.

SOURCES / CLICKABLE REFERENCES

[1] NYC Parks. 2015 Street Tree Census – Tree Data. NYC Open Data, uvpi-gqnh. Collection: 2015–2016. Starting subset count: my Assignment 2 map.

[2] Roman, L. A., J. J. Battles, and J. R. McBride. 2016. *Urban Tree Mortality: A Primer on Demographic Approaches*. USDA Forest Service, NRS-158, p. 9, eq. 10. DOI: 10.2737/NRS-GTR-158.

[3] Hilbert, D. R., L. A. Roman, A. K. Koeser, J. Vogt, and N. S. van Doorn. 2019. “Urban Tree Mortality: A Literature Review.” *Arboriculture & Urban Forestry* 45(5): 167–200, especially pp. 167 and 174.

[4] Vosburgh, A. 2026. “Simulating Trees.” Tutorial 3, especially Steps 1 and 10. *Simulations, Models, Twins*.